



FCC RADIO TEST REPORT

FCC ID : 2AEIM-1470138
Equipment : Wall Connector 3
Brand Name : Tesla
Model Name : 1470138-XX-Y
Applicant : Tesla, Inc.
3500 DEER CREEK ROAD PALO ALTO, CA 94304
Manufacturer : Tesla, Inc.
3500 DEER CREEK ROAD PALO ALTO, CA 94304
Standard : FCC Part 15 Subpart C §15.225

The product was received on Jul. 18, 2023 and testing was performed from Jul. 25, 2023 to Jul. 27, 2023. We, Sporton International (USA) Inc., would like to declare that the tested sample has been evaluated in accordance with the test procedures and has been in compliance with the applicable technical standards.

The test results in this report apply exclusively to the tested model / sample. Without written approval from Sporton International (USA) Inc., the test report shall not be reproduced except in full.

Approved by: Abi Lin

Sporton International (USA) Inc.
1175 Montague Expressway, Milpitas, CA 95035



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History of this test report

Report No.	Version	Description	Issue Date
FR230726002A	01	Initial issue of report	Aug. 16, 2023
FR230726002A	02	Revise model name, Section 1.1 and Section 2.1 This report is an updated version, replacing the report issued on Aug. 16, 2023.	Sep. 28, 2023

Summary of Test Result

Report Clause	Ref Std. Clause	Test Items	Result (PASS/FAIL)	Remark
3.1	15.207	AC Power Line Conducted Emissions	Pass	8.10 dB under the limit at 27.12MHz
3.2	15.215(c)	20dB Spectrum Bandwidth	Pass	-
	2.1049	99% OBW Spectrum Bandwidth	Reporting only	-
3.3	15.225(e)	Frequency Stability	Pass	-
3.4	15.225(a)(b)(c)	Field Strength of Fundamental Emissions	Pass	Max level 34.51 dB μ V/m at 13.56 MHz
3.5	15.225(d) 15.209	Radiated Spurious Emissions	Pass	0.96 dB under the limit at 40.67MHz
3.6	15.203	Antenna Requirements	Pass	-

Conformity Assessment Condition:

1. The test results (PASS/FAIL) with all measurement uncertainty excluded are presented against the regulation limits or in accordance with the requirements stipulated by the applicant/manufacture who shall bear all the risks of non-compliance that may potentially occur if measurement uncertainty is taken into account.
2. The measurement uncertainty please refer to each test result in the section "Measurement Uncertainty".

Disclaimer:

The product specifications of the EUT presented in the test report that may affect the test assessments are declared by the manufacturer who shall take full responsibility for the authenticity.



1. General Description

1.1 Product Feature of Equipment Under Test

Product Feature
General Specs NFC
Antenna Type NFC: Loop Antenna

Remark: The EUT's information above is declared by manufacturer. Please refer to Disclaimer in report summary.

1.2 Modification of EUT

No modifications made to the EUT during the testing.

1.3 Testing Location

Test Site	Sporton International (USA) Inc.		
Test Site Location	1175 Montague Expressway, Milpitas, CA 95035 TEL: (408) 904-3300		
Test Site No.	Sporton Site No.		
	TH01-CA	CO01-CA	03CH01-CA
Test Engineer	Liliana Gonzalez	Fu Chen	Fu Chen
Temperature	24.1 °C	20.2~23.8 °C	23.4~24.1 °C
Relative Humidity	48.0 %	41.2~45.8 %	45.1~51.2 %

Note: The test site complies with ANSI C63.4 2014 requirement.

FCC Designation No.: US1250

1.4 Applicable Standards

According to the specifications declared by the manufacturer, the EUT must comply with the requirements of the following standards:

- ♦ FCC Part 15 Subpart C §15.225
- ♦ FCC KDB 414788 D01 Radiated Test Site v01r01
- ♦ ANSI C63.10-2013

Remark:

1. All the test items were validated and recorded in accordance with the standards without any modification during the testing.
2. This EUT has also been tested and complied with the requirements of FCC Part 15, Subpart B, recorded in a separate test report.

2. Test Configuration of Equipment Under Test

2.1 Descriptions of Test Mode

Investigation has been done on all the possible configurations.

The following table is a list of the test modes shown in this test report.

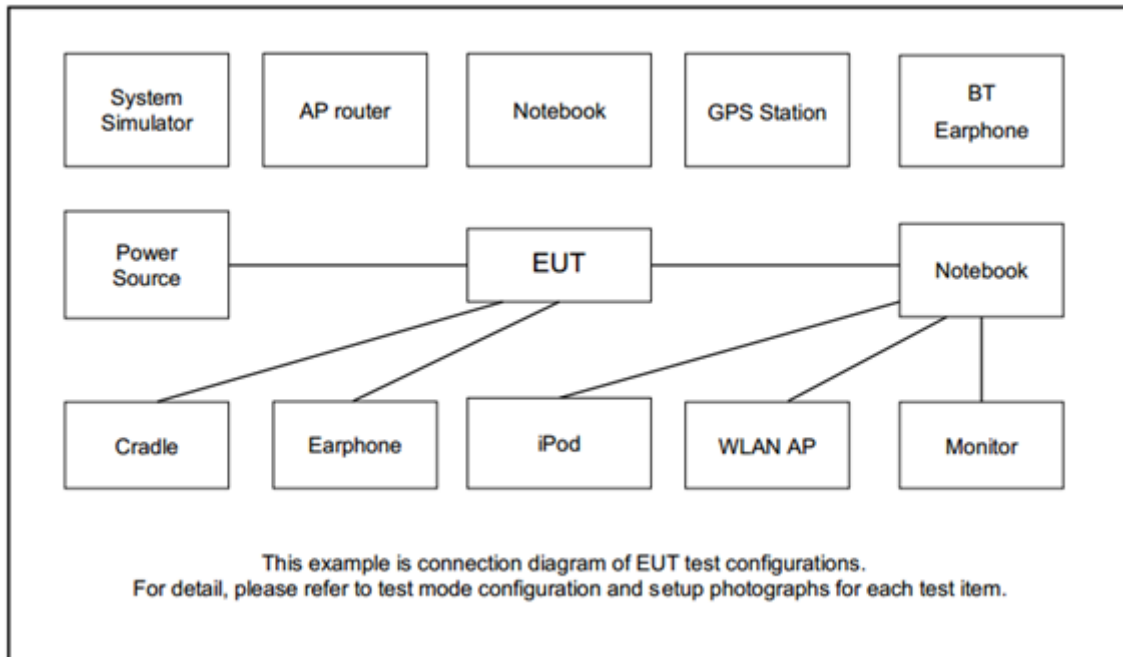
Test Items	
AC Power Line Conducted Emissions	Field Strength of Fundamental Emissions
20dB Spectrum Bandwidth	Frequency Stability
Radiated Emissions 9kHz~30MHz	Radiated Emissions 30MHz~1GHz

The EUT pre-scanned in reader mode with NFC tag (NFC type A) and without reading tag. Based on the highest field strength of fundamental and spurious emissions, the worst case type (type A) was recorded in this report.

The measured emission level of the EUT was maximized by rotating the EUT on a turntable, adjusting the orientation of the EUT and EUT antenna in three orthogonal axis (X: flat, Y: portrait, Z: landscape), and adjusting the measurement antenna orientation, following C63.10 exploratory test procedures and only the worst case emissions were reported in this report.

Test Cases	
AC Conducted Emission	Mode 1: NFC Link + AC power Mode 2: NFC Link + Terminal + AC power
Remark: The worst case of Conducted Emission is mode 1; only the test data of it was reported.	

2.2 Connection Diagram of Test System



2.3 Table for Supporting Units

Item	Equipment	Brand Name	Model Name	FCC ID	Data Cable	Power Cord
1.	NFC Type A Card	N/A	N/A	N/A	N/A	N/A

2.4 EUT Operation Test Setup

The EUT is programmed to be in continuously transmitting mode.

The ancillary equipment, NFC card, is used to make the EUT (NFC) continuously transmitting signal (Power Level: Default) at 13.56MHz and is placed around 0 cm gap to the EUT.

3. Test Results

3.1 AC Power Line Conducted Emissions Measurement

3.1.1 Limit of AC Conducted Emission

For equipment that is designed to be connected to the public utility (AC) power line, the radio frequency voltage that is conducted back onto the AC power line on any frequency or frequencies within the band 150 kHz to 30 MHz shall not exceed the limits in the following table.

Frequency of Emission (MHz)	Conducted Limit (dB μ V)	
	Quasi-Peak	Average
0.15-0.5	66 to 56*	56 to 46*
0.5-5	56	46
5-30	60	50

*Decreases with the logarithm of the frequency.

For terminal test result, the testing follows FCC KDB 174176.

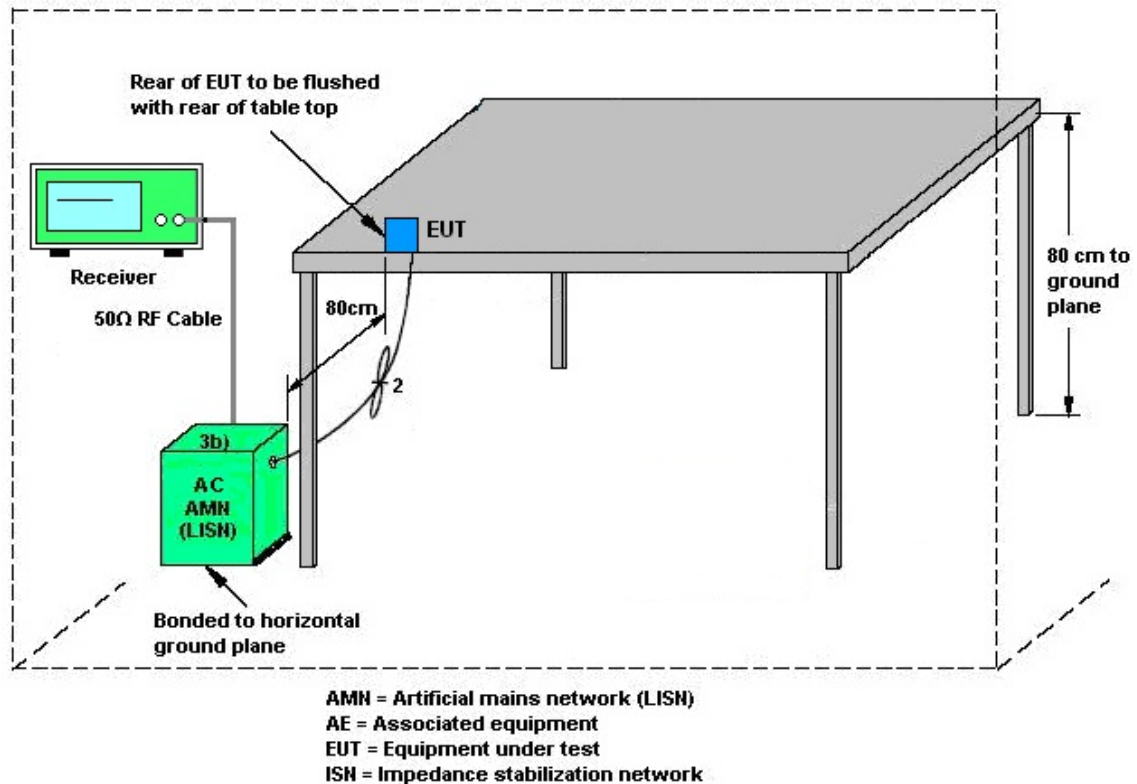
3.1.2 Measuring Instruments

Please refer to the measuring equipment list in this test report.

3.1.3 Test Procedures

1. The EUT is placed 0.4 meter away from the conducting wall of the shielding room, and is kept at least 80 centimeters from any other grounded conducting surface.
2. Connect EUT to the power mains through a line impedance stabilization network (LISN).
3. All the support units are connecting to the other LISN.
4. The LISN provides 50 ohm coupling impedance for the measuring instrument.
5. The FCC states that a 50 ohm, 50 microhenry LISN shall be used.
6. Both Line and Neutral shall be tested in order to find out the maximum conducted emission.
7. The frequency range from 150 kHz to 30 MHz is scanned.
8. Set the test-receiver system to Peak Detect Function and specified bandwidth (IF Bandwidth = 9 kHz) with Maximum Hold Mode. Then measurement is also conducted by Average Detector and Quasi-Peak Detector Function respectively.

3.1.4 Test setup



3.1.5 Test Result of AC Conducted Emission

Please refer to Appendix A.

Note:

(1) with antenna

Remark: 13.5XXMHz is the NFC RF fundamental signal.

(2) with dummy load

Remark: Only the fundamental NFC signal needs to be retested per C63.4.

3.2 20dB and 99% OBW Spectrum Bandwidth Measurement

3.2.1 Limit

Intentional radiators must be designed to ensure that the 20 dB and 99% emission bandwidth in the specific band 13.553~13.567 MHz.

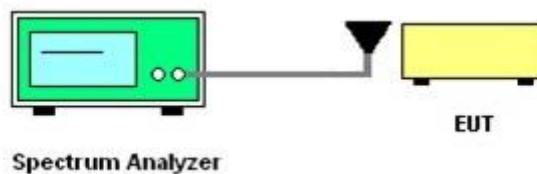
3.2.2 Measuring Instruments

Please refer to the measuring equipment list in this test report.

3.2.3 Test Procedures

1. The spectrum analyzer connected via a receive antenna placed near the EUT in peak Max Hold Mode.
2. The resolution bandwidth of 1 kHz and the video bandwidth of 3 kHz were used.
3. Measured the spectrum width with power higher than 20 dB below carrier.
4. Measured the 99% OBW.

3.2.4 Test Setup



3.2.5 Test Result of Near Field Test Items

Please refer to Appendix B.

3.3 Frequency Stability Measurement

3.3.1 Limit

The frequency tolerance of the carrier signal shall be maintained within +/- 0.01% (100ppm) of the operating frequency over a temperature variation of -20 degrees to +50 degrees C at normal supply voltage, and for a variation in the primary supply voltage from 85% to 115% of the rated supply voltage at a temperature of 20 degrees C. For battery operated equipment, the equipment tests shall be performed by using a new battery.

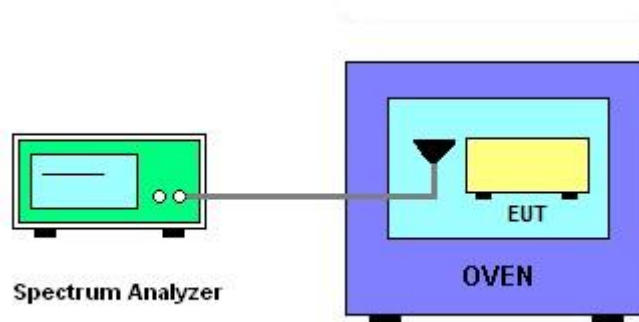
3.3.2 Measuring Instruments

Please refer to the measuring equipment list in this test report.

3.3.3 Test Procedures

1. The spectrum analyzer connected via a receive antenna placed near the EUT.
2. EUT has transmitted signal and fixed channelize.
3. Set the spectrum analyzer span to view the entire emissions bandwidth.
4. Set RBW = 1 kHz, VBW = 3 kHz with peak detector and maxhold settings.
5. The f_c is declaring of channel frequency. Then the frequency error formula is $(f_c - f)/f_c \times 10^6$ ppm and the limit is less than ± 100 ppm.
6. Extreme temperature rule is -20°C~50°C.

3.3.4 Test Setup



3.3.5 Test Result of Near Field Test Items

Please refer to Appendix B.

3.4 Field Strength of Fundamental Emissions and Mask Measurement

3.4.1 Limit

Rules and specifications	FCC CFR 47 Part 15 section 15.225			
Description	Compliance with the spectrum mask is tested with RBW set to 9kHz.			
Freq. of Emission (MHz)	Field Strength (μ V/m) at 30m	Field Strength (dB μ V/m) at 30m	Field Strength (dB μ V/m) at 10m	Field Strength (dB μ V/m) at 3m
1.705~13.110	30	29.5	48.58	69.5
13.110~13.410	106	40.5	59.58	80.5
13.410~13.553	334	50.5	69.58	90.5
13.553~13.567	15848	84.0	103.08	124.0
13.567~13.710	334	50.5	69.58	90.5
13.710~14.010	106	40.5	59.58	80.5
14.010~30.000	30	29.5	48.58	69.5

Remark:

1. The field strength test result is in 3m test distance, follow test rules the test data use distance extrapolation factor and reported in this report at 30m test result.
2. Distance extrapolation factor = $40 \log (\text{specific distance} / \text{test distance})$ (dB)

3.4.2 Measuring Instruments

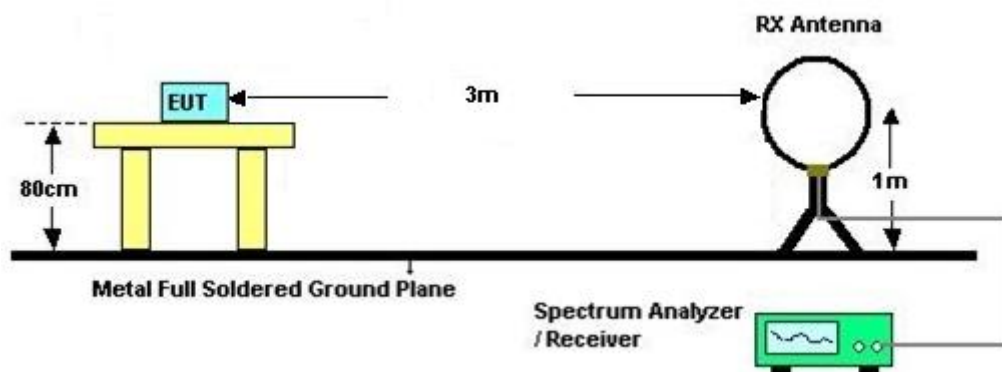
Please refer to the measuring equipment list in this test report.

3.4.3 Test Procedures

1. Configure the EUT according to ANSI C63.10. The EUT is placed on the top of the turntable 0.8 meter above ground. The phase center of the loop receiving antenna mounted antenna tower is placed 3 meters far away from the turntable.
 2. Power on the EUT and all the supporting units. The turntable is rotated by 360 degrees to determine the position of the highest radiation.
 3. The height of the receiving antenna is fixed at one meter above ground to find the maximum emissions field strength.
 4. For Fundamental emissions, use the receiver to measure QP reading.
 5. When the radiated emissions limits are expressed in terms of the average value of the emissions, and pulsed operation is employed, the measurement field strength shall be determined by averaging over one complete pulse train, including blanking intervals, as long as the pulse train does not exceed 0.1 seconds. As an alternative (provided the transmitter operates for longer than 0.1 seconds) or in cases where the pulse train exceeds 0.1 seconds, the measured field strength shall be determined from the average absolute voltage during a 0.1 second interval during which the field strength is at its maximum value.
 6. Compliance with the spectrum mask is tested with RBW set to 9 kHz.
- Note: Emission level (dBμV/m) = 20 log Emission level (μV/m).

3.4.4 Test Setup

For radiated test below 30MHz



3.4.5 Test Result of Field Strength of Fundamental Emissions and Mask

Please refer to Appendix C.

3.5 Radiated Emissions Measurement

3.5.1 Limit

The field strength of any emissions which appear outside of 13.110 ~14.010MHz band shall not exceed the general radiated emissions limits.

Frequencies (MHz)	Field Strength (μ V/m)	Measurement Distance (meters)
0.009~0.490	2400/F(kHz)	300
0.490~1.705	24000/F(kHz)	30
1.705~30.0	30	30
30~88	100	3
88~216	150	3
216~960	200	3
Above 960	500	3

3.5.2 Measuring Instruments

Please refer to the measuring equipment list in this test report.

3.5.3 Measuring Instrument Setting

The following table is the setting of receiver:

Receiver Parameter	Setting
Attenuation	Auto
Frequency Range: 9kHz~150kHz	RBW 200Hz for QP
Frequency Range: 150kHz~30MHz	RBW 9kHz for QP
Frequency Range: 30MHz~1000MHz	RBW 120kHz for Peak

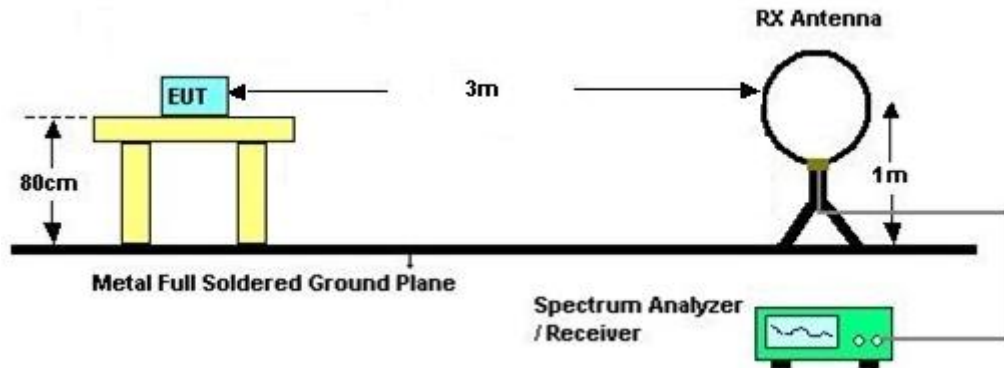
Note: The emission limits shown in the above table are based on measurements employing a CISPR quasi-peak detector except for the frequency bands 9-90 kHz and 110-490 kHz. Radiated emission limits in these two bands are based on measurements employing an average detector.

3.5.4 Test Procedures

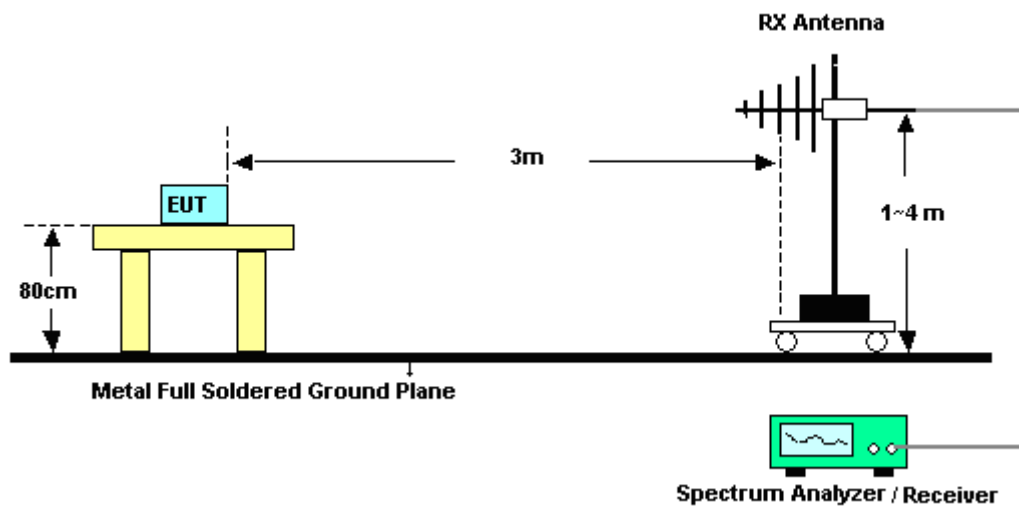
1. Configure the EUT according to ANSI C63.10. The EUT is placed on the top of the turntable 0.8 meter above ground. The phase center of the receiving antenna mounted on the top of a height-variable antenna tower is placed 3 meters far away from the turntable.
2. Power on the EUT and all the supporting units. The turntable is rotated by 360 degrees to determine the position of the highest radiation.
3. The height of the broadband receiving antenna is varied between one meter and four meters above ground to find the maximum emissions field strength of both horizontal and vertical polarization.
4. For each suspected emissions, the antenna tower is scanned (from 1 M to 4 M) and then the turntable is rotated (from 0 degree to 360 degrees) to find the maximum reading.
5. Set the test-receiver system to Peak or CISPR quasi-peak Detect Function with specified bandwidth under Maximum Hold Mode.
6. When the radiated emissions limits are expressed in terms of the average value of the emissions, and pulsed operation is employed, the measurement field strength shall be determined by averaging over one complete pulse train, including blanking intervals, as long as the pulse train does not exceed 0.1 seconds. As an alternative (provided the transmitter operates for longer than 0.1 seconds) or in cases where the pulse train exceeds 0.1 seconds, the measured field strength shall be determined from the average absolute voltage during a 0.1 second interval during which the field strength is at its maximum value.
7. In case the emission is lower than 30 MHz, loop antenna has to be used for measurement and the recorded data shall be QP measured by receiver.
8. Radiated testing below 1 GHz is performed by adjusting the antenna tower from 1 m to 4 m and by rotating the turn table from 0 degree to 360 degrees to find the peak maximum hold reading. When there is no suspected emission found and the emission level is with at least 6 dB margin against QP limit line, the position is marked as “-”.

3.5.5 Test Setup

For radiated test below 30MHz



For radiated test above 30MHz



3.5.6 Test Result of Radiated Emissions Measurement

Please refer to Appendix C.

Remark: There is adequate comparison measurement of both open-field test site and alternative test site - semi-Anechoic chamber according to 414788 D01 Radiated Test Site v01r01, and the result came out very similar.

3.6 Antenna Requirements

3.6.1 Standard Applicable

Except for special regulations, the Low-power Radio-frequency Devices must not be equipped with any jacket for installing an antenna with extension cable. An intentional radiator shall be designed to ensure that no antenna other than that furnished by the responsible party shall be used with the device. The use of a permanently attached antenna or of an antenna that uses a unique coupling to the intentional radiator shall be considered sufficient to comply with the provisions of this Section. The manufacturer may design the unit so that the user can replace a broken antenna, but the use of a standard antenna jack or electrical connector is prohibited.

The use of a permanently attached antenna or of an antenna that uses a unique coupling to the intentional radiator shall be considered sufficient to comply with the rule.

3.6.2 Antenna Anti-Replacement Construction

An embedded-in antenna design is used.



4. List of Measuring Equipment

Instrument	Brand Name	Model No.	Serial No.	Characteristics	Calibration Date	Test Date	Due Date	Remark
LISN	TESEQ	NNB51	47407	N/A	May 16, 2023	Jul. 27, 2023	May 15, 2024	Conduction (CO01-CA)
EMI Test Receiver	R&S	ESR7	102177	9kHz~7GHz	May 23, 2023	Jul. 27, 2023	May 22, 2024	Conduction (CO01-CA)
Pulse limiter with 10dB attenuation	R&S	VTSD 9561-F N	9561-F-N00412	N/A	Jun. 05, 2023	Jul. 27, 2023	Jun. 04, 2024	Conduction (CO01-CA)
Test Software	R&S	EMC32 V10.30.0	N/A	N/A	N/A	Jul. 27, 2023	N/A	Conduction (CO01-CA)
Hygrometer	Testo	608-H1	45142602	N/A	Sep. 12, 2022	Jul. 25, 2023	Sep. 11, 2023	Conducted (TH01-CA)
Spectrum Analyzer	Rohde & Schwarz	FSV40	101545	10Hz-40GHz	May 03, 2023	Jul. 25, 2023	May 02, 2024	Conducted (TH01-CA)
Temperature & Humidity Chamber	ESPEC	SH-642	93012171	N/A	Sep. 06, 2022	Jul. 25, 2023	Sep. 05, 2023	Conducted (TH01-CA)
AC Power Source	AC Power Corp.	AFC-11003F	F319020053	Output: 5~300V	N/A	Jul. 25, 2023	N/A	Conducted (TH01-CA)
Bilog Antenna	TESEQ	6111D	54683	30MHz~1GHz	Nov. 01, 2022	Jul. 27, 2023	Oct. 31, 2023	Radiation (03CH01-CA)
Loop Antenna	R&S	HFH2-Z2E	100840	9kHz~30MHz	Jun. 29, 2023	Jul. 27, 2023	Jun. 28, 2024	Radiation (03CH01-CA)
Preamplifier	SONOMA	310N	372241	9kHz~1GHz	May 03, 2023	Jul. 27, 2023	May 02, 2024	Radiation (03CH01-CA)
EMI Test Receiver	R&S	ESU26	100049	20Hz~26.5GHz	May 02, 2023	Jul. 27, 2023	May 01, 2024	Radiation (03CH01-CA)
RF Cable	HUBER+SU HNER	SUCOFLEX 102	8015932/2, 8015762/2, 804938/2	N/A	Mar. 06, 2023	Jul. 27, 2023	Mar. 05, 2024	Radiation (03CH01-CA)
Filter	Wainwright	WHK20/1000C7 /40SS	SN1	20MHz High Pass Filter	Jun. 05, 2023	Jul. 27, 2023	Jun. 04, 2024	Radiation (03CH01-CA)
Hygrometer	TESTO	608-H1	45142559	N/A	Sep. 12, 2022	Jul. 27, 2023	Sep. 11, 2023	Radiation (03CH01-CA)
Controller	Chaintek	EM-1000	060881	Control Turn Table & Antenna Mast	N/A	Jul. 27, 2023	N/A	Radiation (03CH01-CA)
Antenna Mast	ChainTek	MBS-520-1	N/A	1m~4m	N/A	Jul. 27, 2023	N/A	Radiation (03CH01-CA)
Test Software	Audix E3	E6.2009-8-24d	PK-002093	N/A	N/A	Jul. 27, 2023	N/A	Radiation (03CH01-CA)

5. Measurement Uncertainty

Uncertainty of Conducted Emission Measurement (150 kHz ~ 30 MHz)

Measuring Uncertainty for a Level of Confidence of 95% ($U = 2U_c(y)$)	2.7 dB
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Uncertainty of Radiated Emission Measurement (9 kHz ~ 30 MHz)

Measuring Uncertainty for a Level of Confidence of 95% ($U = 2U_c(y)$)	3.7 dB
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Uncertainty of Radiated Emission Measurement (30 MHz ~ 1000 MHz)

Measuring Uncertainty for a Level of Confidence of 95% ($U = 2U_c(y)$)	4.6 dB
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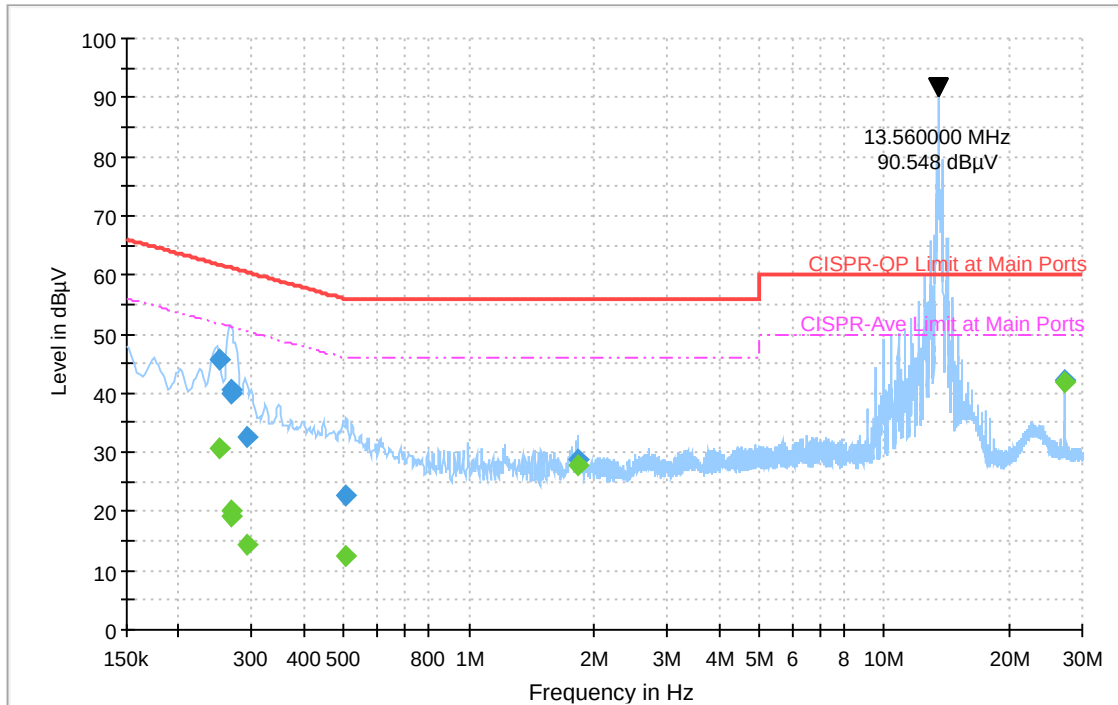
Appendix A. Test Results of Conducted Emission Test

<Original>

EUT Information

Site: CO01-CA
Power: 220Vac/60Hz
Project: 230726002

Full Spectrum



Final Result

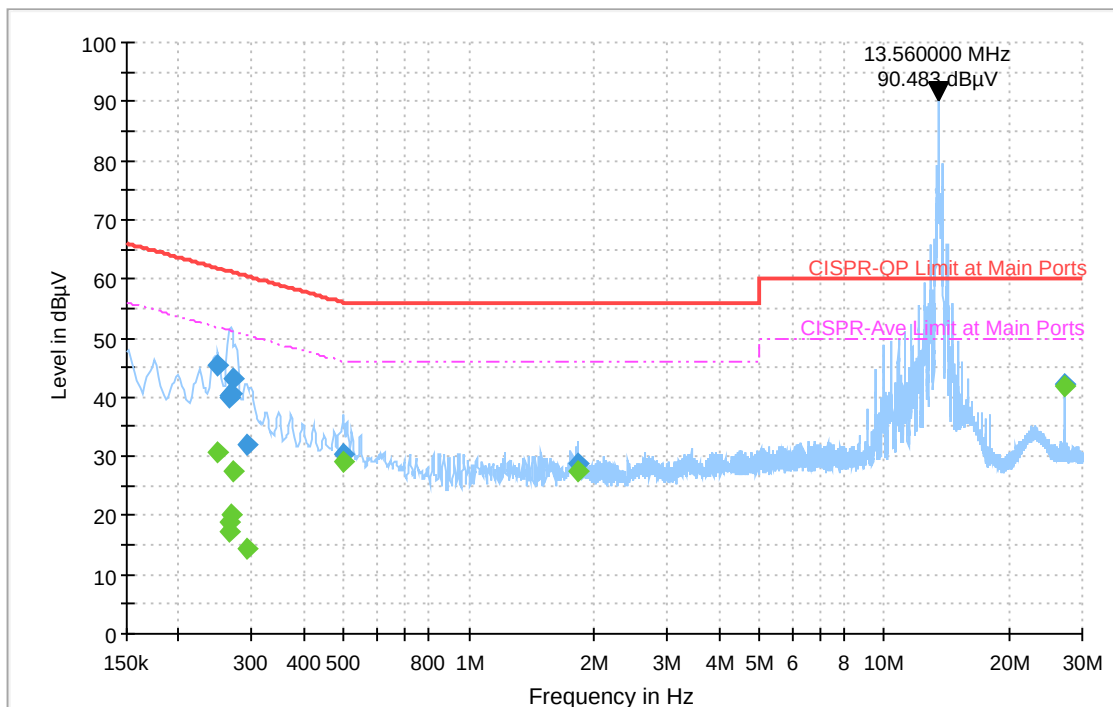
Frequency (MHz)	QuasiPeak (dBμV)	CAverage (dBμV)	Limit (dBμV)	Margin (dB)	Line	Filter	Corr. (dB)
0.249459	---	30.67	51.78	21.11	L1	OFF	20.3
0.249459	45.58	---	61.78	16.20	L1	OFF	20.3
0.266406	39.93	---	61.23	21.30	L1	OFF	20.3
0.266406	---	19.14	51.23	32.09	L1	OFF	20.3
0.266757	40.47	---	61.22	20.75	L1	OFF	20.3
0.266757	---	20.22	51.22	31.00	L1	OFF	20.3
0.290976	---	14.46	50.50	36.04	L1	OFF	20.3
0.290976	32.68	---	60.50	27.82	L1	OFF	20.3
0.506949	---	12.59	46.00	33.41	L1	OFF	20.3
0.506949	22.75	---	56.00	33.25	L1	OFF	20.3
1.831173	---	27.90	46.00	18.10	L1	OFF	20.3
1.831173	28.76	---	56.00	27.24	L1	OFF	20.3
27.120012	---	41.85	50.00	8.15	L1	OFF	20.7
27.120012	42.13	---	60.00	17.87	L1	OFF	20.7

EUT Information

Site:
Power:
Project

CO01-CA
220Vac/60Hz
230726002

Full Spectrum



Final Result

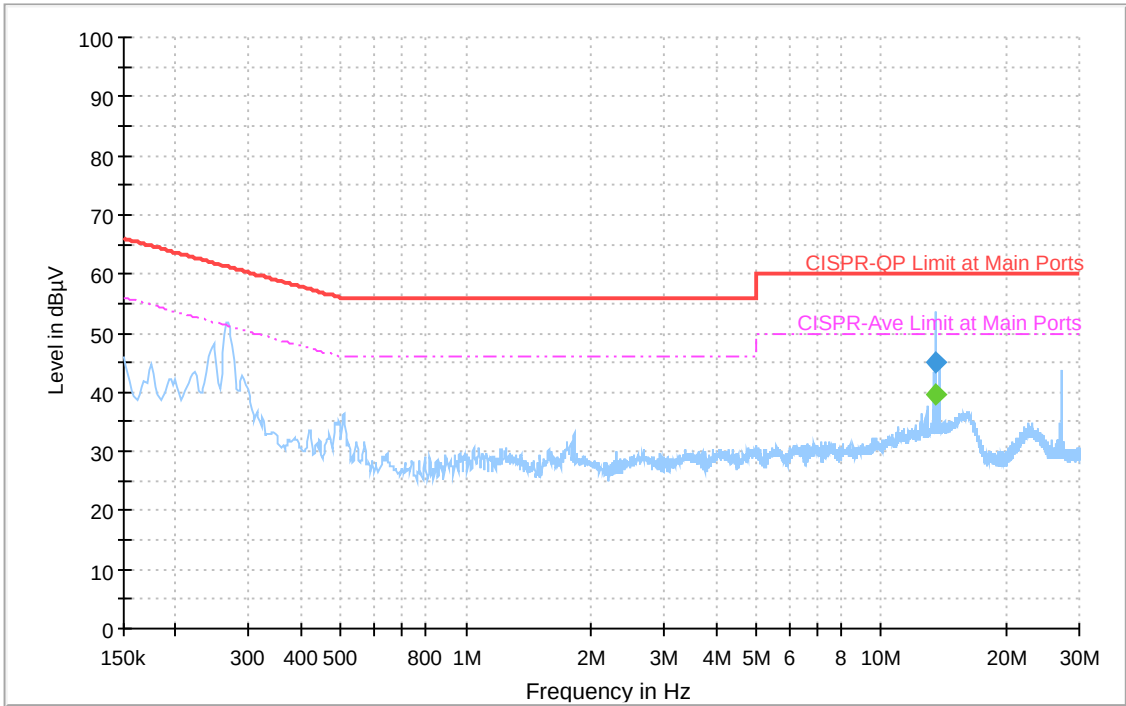
Frequency (MHz)	QuasiPeak (dBμV)	CAverage (dBμV)	Limit (dBμV)	Margin (dB)	Line	Filter	Corr. (dB)
0.248766	45.50	---	61.80	16.30	N	OFF	20.3
0.248766	---	30.55	51.80	21.25	N	OFF	20.3
0.265542	---	17.39	51.26	33.87	N	OFF	20.3
0.265542	40.00	---	61.26	21.26	N	OFF	20.3
0.266109	---	18.75	51.24	32.49	N	OFF	20.3
0.266109	40.31	---	61.24	20.93	N	OFF	20.3
0.267027	40.50	---	61.21	20.71	N	OFF	20.3
0.267027	---	20.19	51.21	31.02	N	OFF	20.3
0.270789	43.25	---	61.09	17.84	N	OFF	20.3
0.270789	---	27.38	51.09	23.71	N	OFF	20.3
0.291255	31.83	---	60.49	28.66	N	OFF	20.3
0.291255	---	14.45	50.49	36.04	N	OFF	20.3
0.497139	30.30	---	56.05	25.75	N	OFF	20.3
0.497139	---	29.06	46.05	16.99	N	OFF	20.3
1.829481	28.62	---	56.00	27.38	N	OFF	20.3
1.829481	---	27.61	46.00	18.39	N	OFF	20.3
27.119805	42.20	---	60.00	17.80	N	OFF	20.7
27.119805	---	41.90	50.00	8.10	N	OFF	20.7

EUT Information

Site:
Power:
Project

CO01-CA
220Vac/60Hz
230726002
board with terminal

Full Spectrum



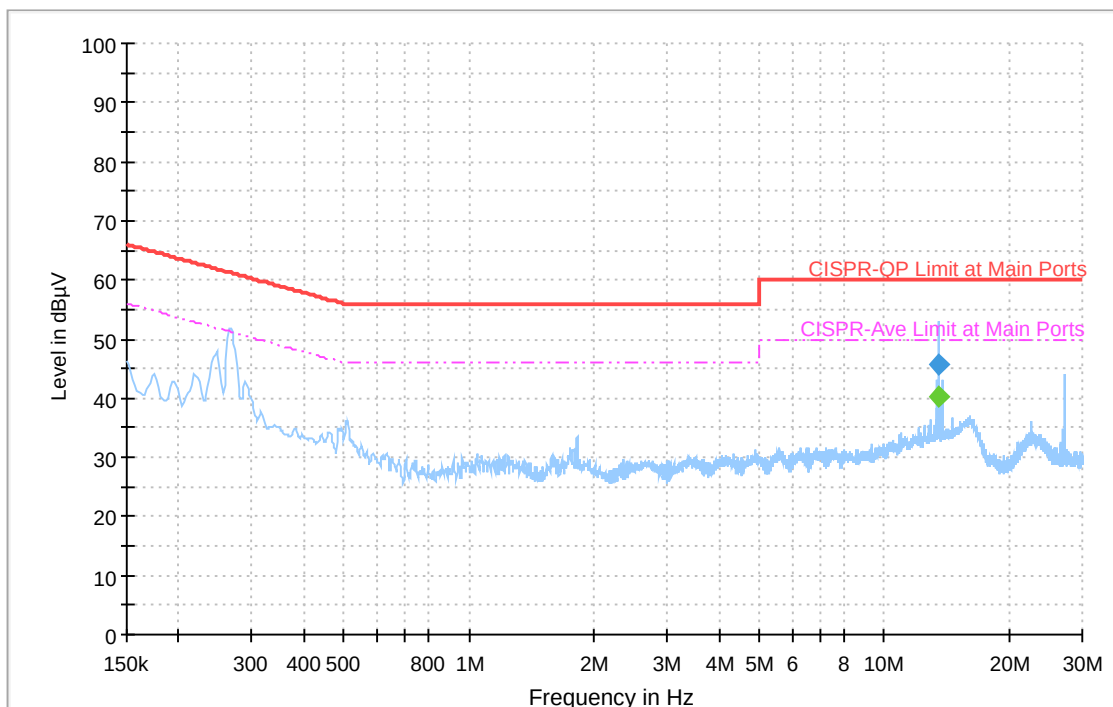
Final Result

Frequency (MHz)	QuasiPeak (dBµV)	CAverage (dBµV)	Limit (dBµV)	Margin (dB)	Line	Filter	Corr. (dB)
13.561782	45.16	---	60.00	14.84	L1	OFF	20.5
13.561782	---	39.50	50.00	10.50	L1	OFF	20.5

EUT Information

Site: CO01-CA
Power: 220Vac/60Hz
Project: 230726002
board with terminal

Full Spectrum



Final Result

Frequency (MHz)	QuasiPeak (dBµV)	CAverage (dBµV)	Limit (dBµV)	Margin (dB)	Line	Filter	Corr. (dB)
13.559712	---	40.12	50.00	9.88	N	OFF	20.5
13.559712	45.71	---	60.00	14.29	N	OFF	20.5



Appendix B. Test Results of Near Field Test Items

B1. Test Result of 20dB Spectrum Bandwidth

Test mode		NFC Tx		Test Frequency (MHz)		13.56																																																									
Spectrum Ref Level 0.00 dBm Att 10 dB RBW 1 kHz SWT 30 ms VBW 3 kHz Mode Sweep IPk Max M1 -16.43 dBm 13.5599800 MHz 20.00 dB 4.416000000 kHz 3070.9 T1 -36.38 dBm 13.557902 MHz 20.00 dB 3070.9 T2 -36.41 dBm 13.562318 MHz 20.00 dB 3070.9 CF 13.56 MHz 1001 pts Span 20.0 kHz Marker <table><thead><tr><th>Type</th><th>Ref</th><th>Trc</th><th>X-value</th><th>Y-value</th><th>Function</th><th>Function Result</th></tr></thead><tbody><tr><td>M1</td><td></td><td>1</td><td>13.55998 MHz</td><td>-16.43 dBm</td><td>ndB down</td><td>4.416 kHz</td></tr><tr><td>T1</td><td></td><td>1</td><td>13.557902 MHz</td><td>-36.38 dBm</td><td>ndB</td><td>20.00 dB</td></tr><tr><td>T2</td><td></td><td>1</td><td>13.562318 MHz</td><td>-36.41 dBm</td><td>Q factor</td><td>3070.9</td></tr></tbody></table>				Type	Ref	Trc	X-value	Y-value	Function	Function Result	M1		1	13.55998 MHz	-16.43 dBm	ndB down	4.416 kHz	T1		1	13.557902 MHz	-36.38 dBm	ndB	20.00 dB	T2		1	13.562318 MHz	-36.41 dBm	Q factor	3070.9	Spectrum Ref Level 0.00 dBm Att 10 dB RBW 1 kHz SWT 30 ms VBW 3 kHz Mode Sweep IPk Max M1 -16.39 dBm 13.5599800 MHz 20.00 dB 13.706293706 kHz T1 -37.99 dBm 13.5531868 MHz 20.00 dB 13.706293706 kHz T2 -38.45 dBm 13.5668931 MHz 20.00 dB 13.706293706 kHz CF 13.56 MHz 1001 pts Span 20.0 kHz Marker <table><thead><tr><th>Type</th><th>Ref</th><th>Trc</th><th>X-value</th><th>Y-value</th><th>Function</th><th>Function Result</th></tr></thead><tbody><tr><td>M1</td><td></td><td>1</td><td>13.55998 MHz</td><td>-16.39 dBm</td><td></td><td></td></tr><tr><td>T1</td><td></td><td>1</td><td>13.5531868 MHz</td><td>-37.99 dBm</td><td>Occ Bw</td><td>13.706293706 kHz</td></tr><tr><td>T2</td><td></td><td>1</td><td>13.5668931 MHz</td><td>-38.45 dBm</td><td></td><td></td></tr></tbody></table>				Type	Ref	Trc	X-value	Y-value	Function	Function Result	M1		1	13.55998 MHz	-16.39 dBm			T1		1	13.5531868 MHz	-37.99 dBm	Occ Bw	13.706293706 kHz	T2		1	13.5668931 MHz	-38.45 dBm		
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Date: 25.JUL.2023 11:42:37				Date: 25.JUL.2023 11:41:39																																																											
20dB Bandwidth (kHz)		4.416		99% OccupiedBW(kHz)		13.706																																																									
Frequency range (MHz)		f _L > 13.553		13.55790		Test Result																																																									
		f _H < 13.567		13.56231		Complies																																																									

Remark: Because the measured signal is CW adjusting the RBW per C63.10 would not be practical since measured bandwidth will always follow the RBW and the result will be approximately twice the RBW.

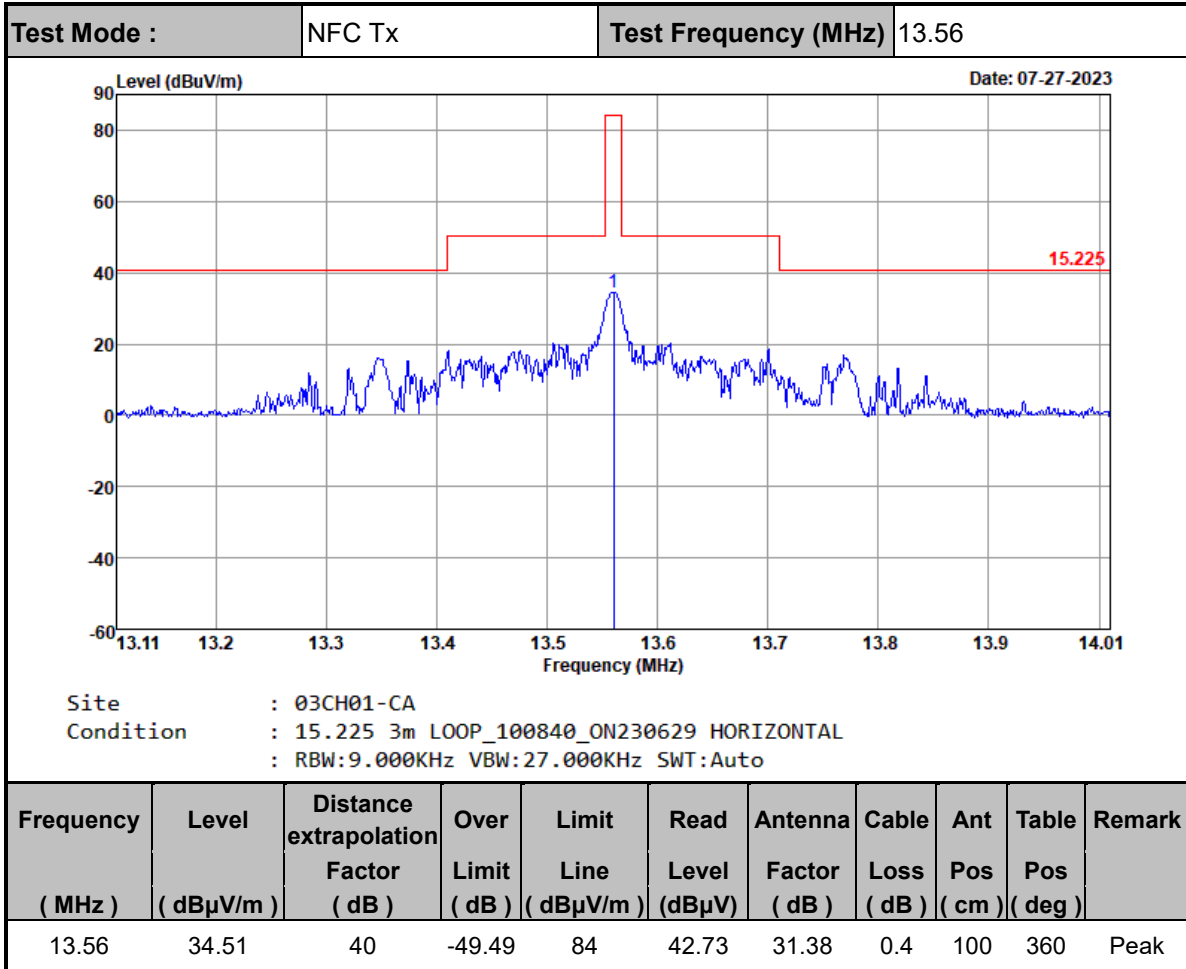
B2. Test Result of Frequency Stability

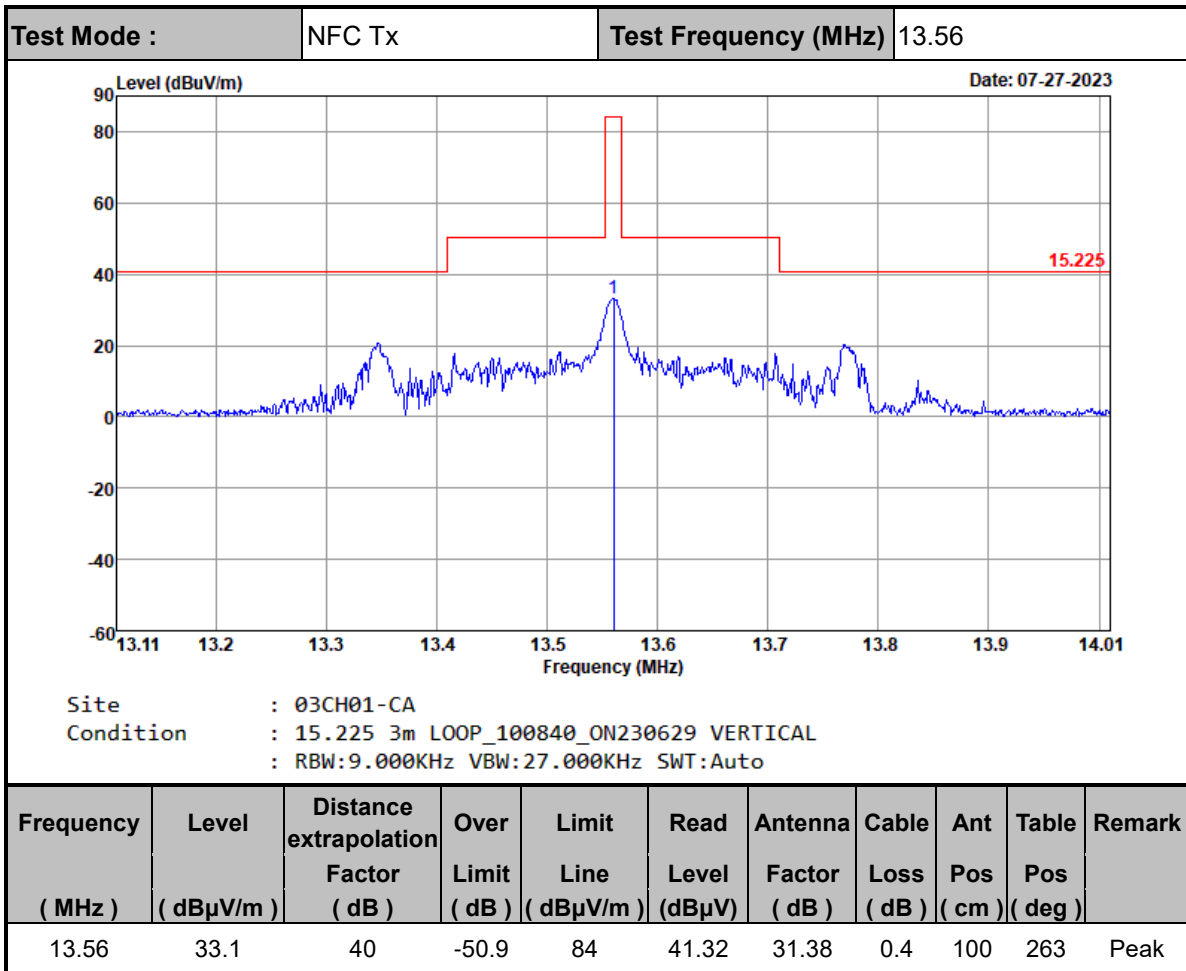
Voltage (Vac)	Measured Frequency (MHz)	Deviation (MHz)	Deviation (ppm)
187	13.560040	0.00004	2.949853
220	13.560000	0.00000	0.000000
253	13.560040	0.00004	2.949853

Temperature (°C)	Measured Frequency (MHz)	Deviation (MHz)	Deviation (ppm)
-20	13.560020	0.000060	4.424779
-10	13.560040	0.000060	4.424779
0	13.560040	0.000060	4.424779
10	13.560020	0.000040	2.949853
20	13.560000	-0.000020	-1.474926
30	13.560040	0.000040	2.949853
40	13.560020	-0.000040	-2.949853
50	13.559980	-0.000020	-1.474926
Limit	FS < ±100 ppm	Limit	FS < ±100 ppm
Test Result	PASS	Test Result	PASS

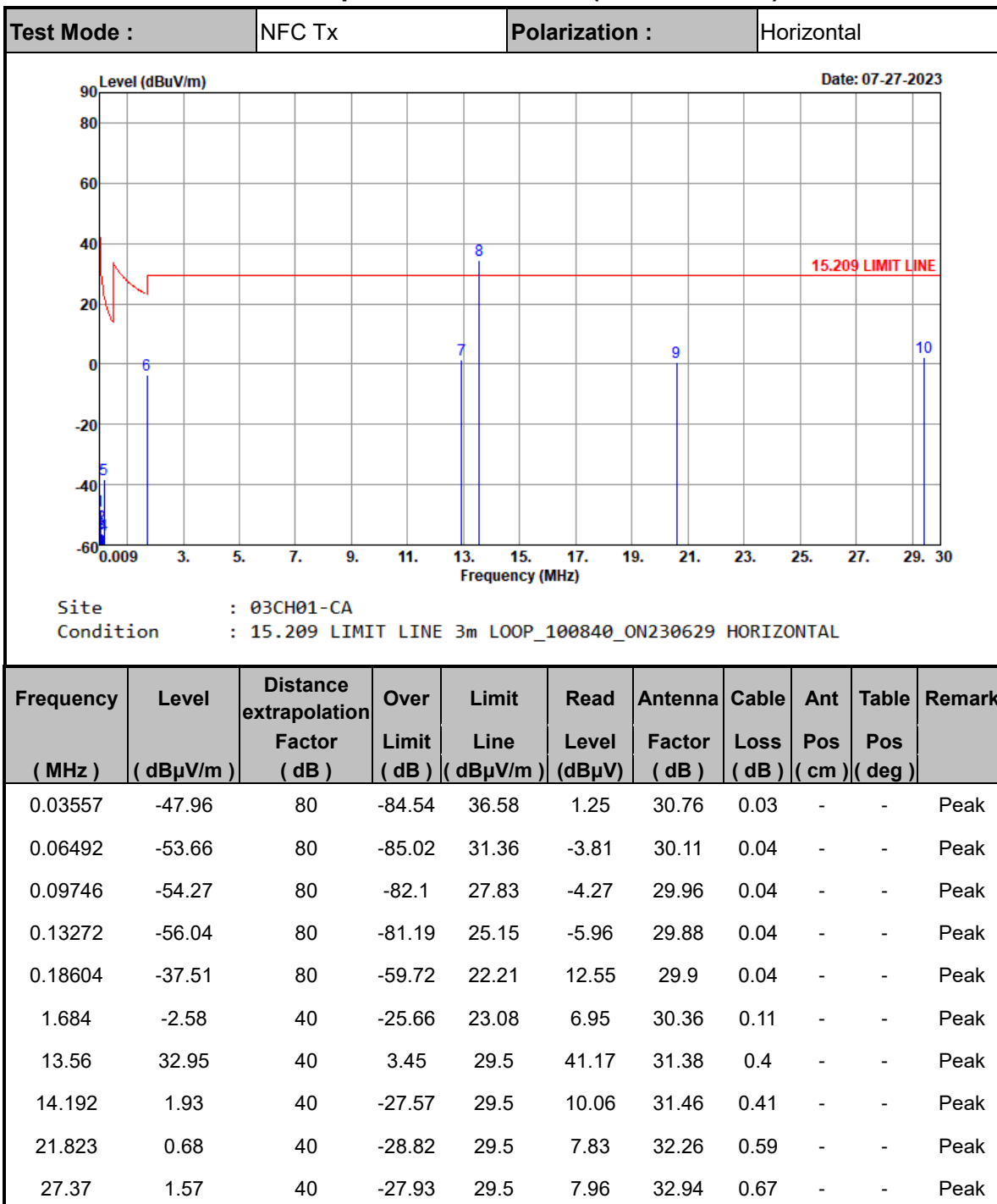
Appendix C. Test Results of Radiated Test Items

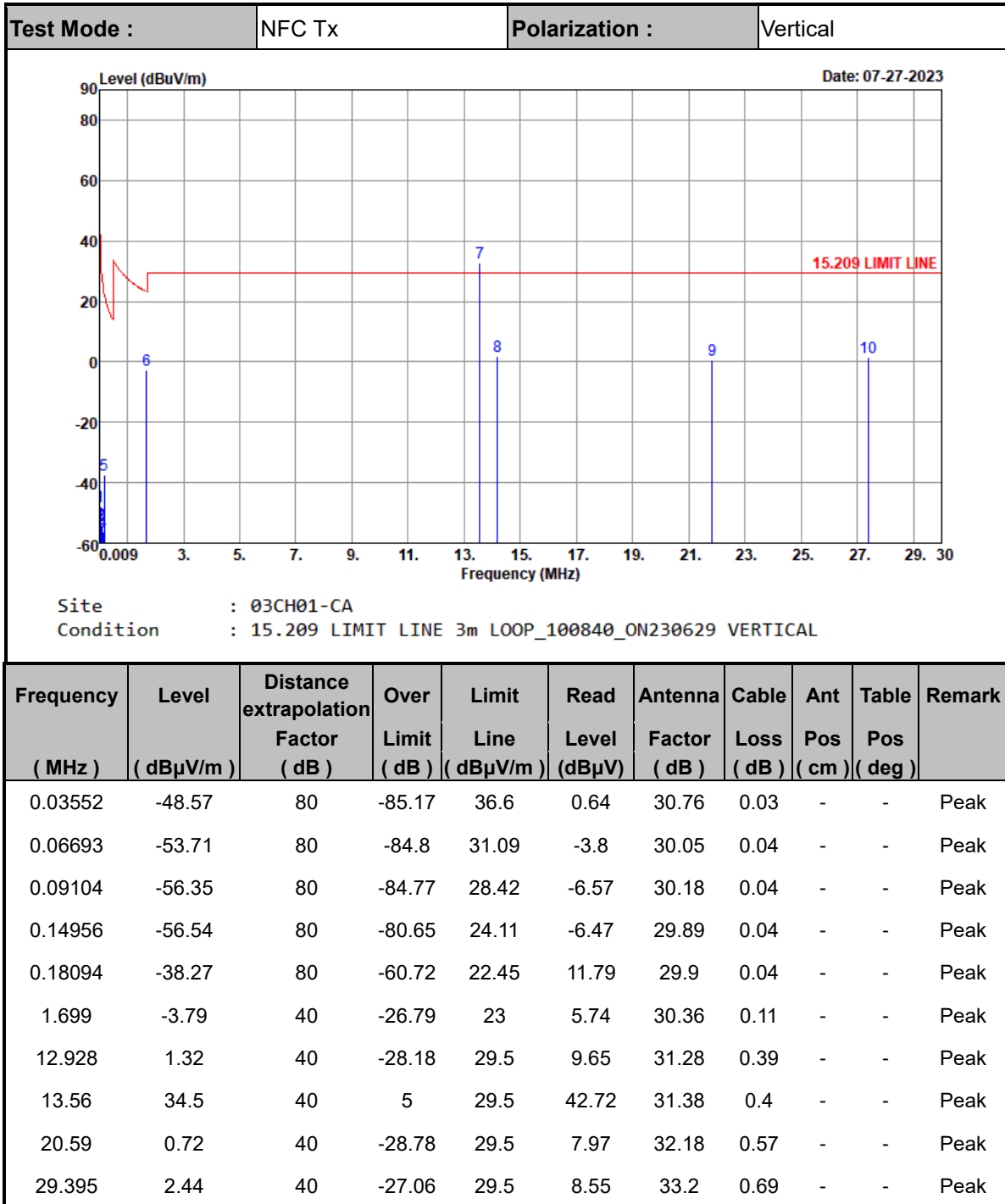
C1. Test Result of Field Strength of Fundamental Emissions



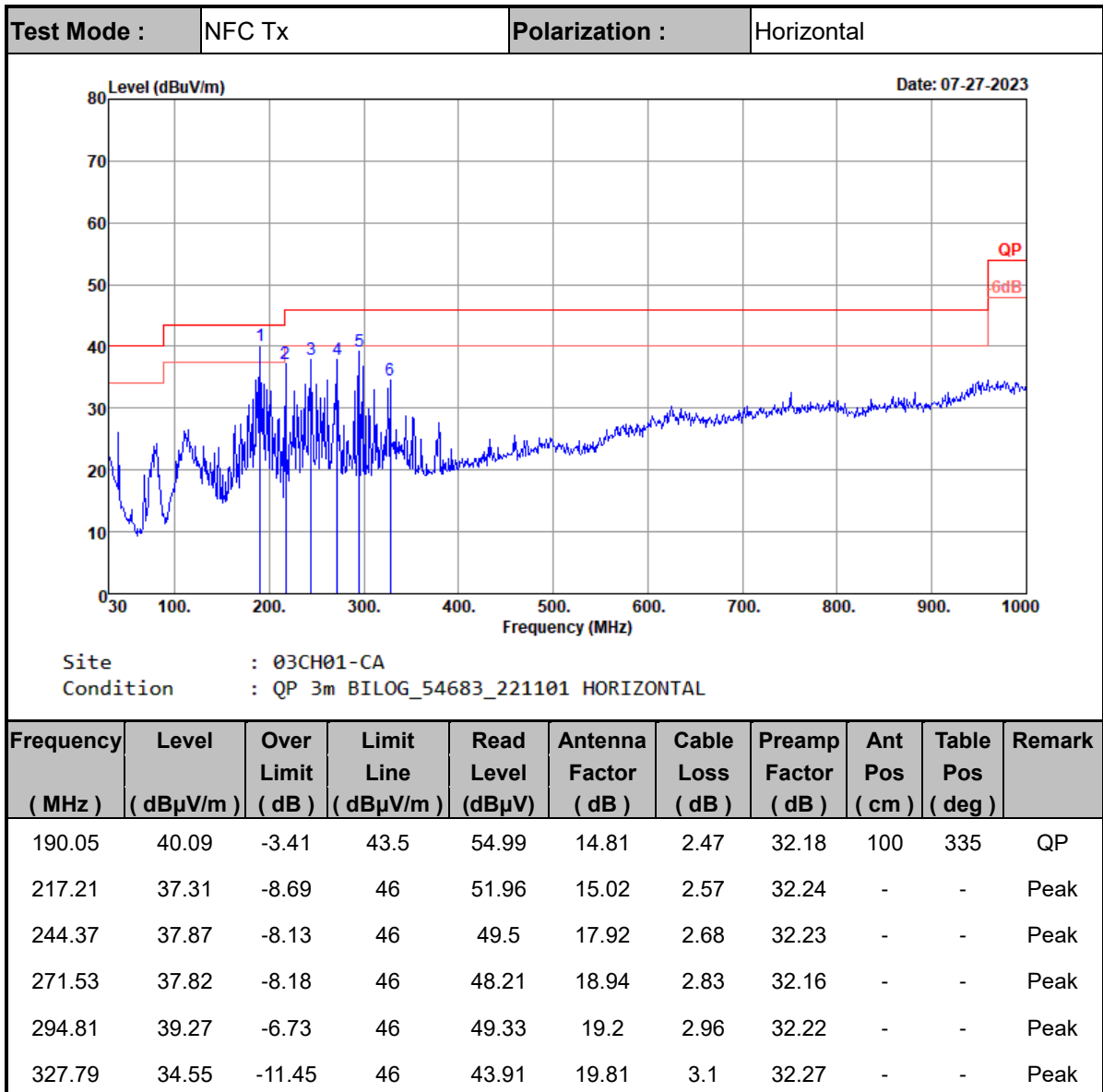

Note :

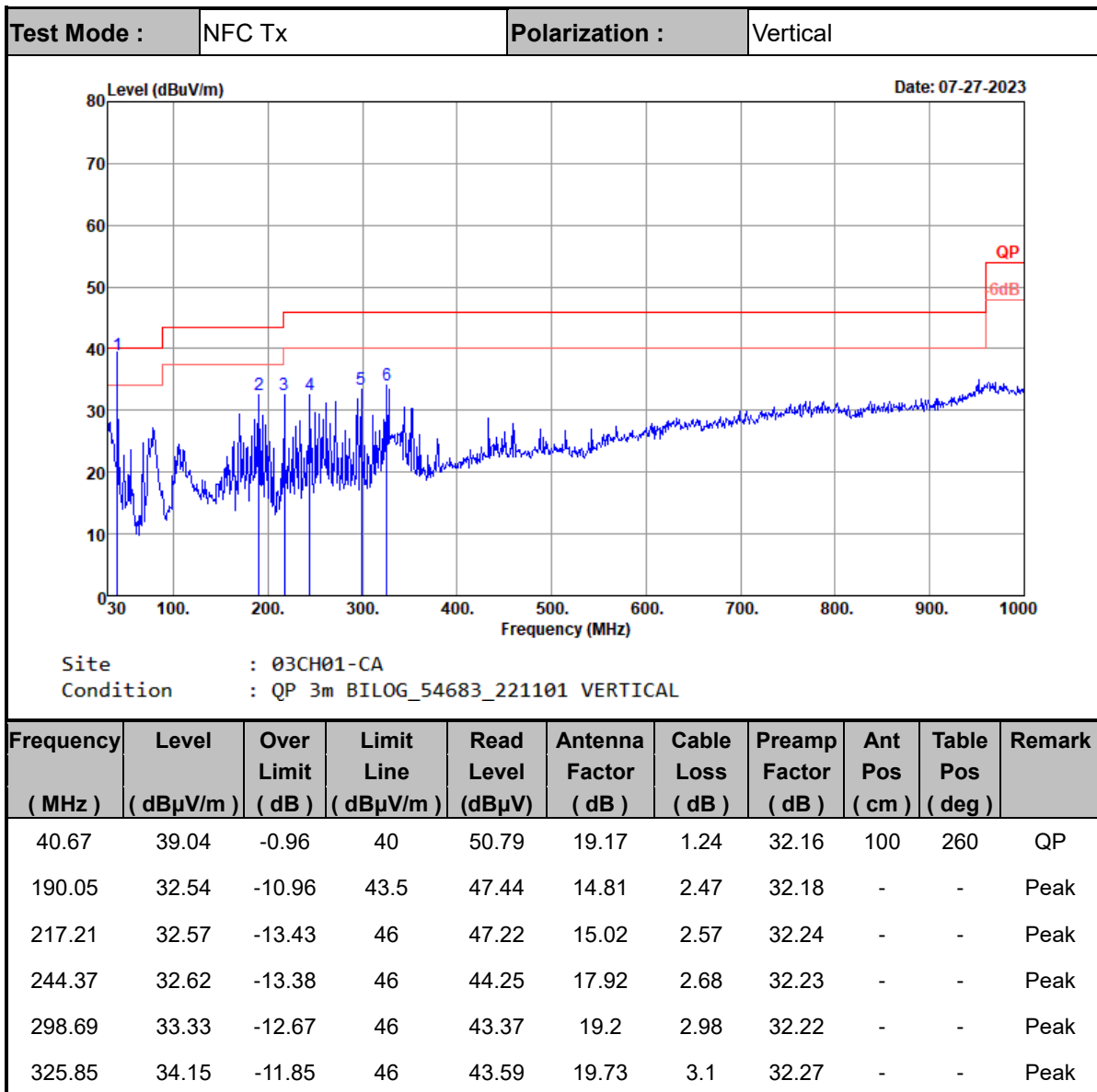
1. Distance extrapolation factor = $40 \log (\text{specific distance} / \text{test distance})$ (dB)
2. Level = Antenna Factor + Cable Loss + Read Level - Distance extrapolation factor.

C2. Results of Radiated Spurious Emissions (9 kHz~30MHz)



Note :

1. The amplitude of spurious emissions which are attenuated by more than 20dB below the permissible value has no need to be reported.
2. Distance extrapolation factor = 40 log (specific distance / test distance) (dB)
3. Level = Antenna Factor + Cable Loss + Read Level - Distance extrapolation factor.
4. 13.56 MHz is fundamental signal which can be ignored

C3. Results of Radiated Spurious Emissions (30MHz~1GHz)



Note:

1. The amplitude of spurious emissions which are attenuated by more than 20dB below the permissible value has no need to be reported.
2. Emission level (dBμV/m) = 20 log Emission level (μV/m).
3. Corrected Reading: Antenna Factor + Cable Loss + Read Level - Preamp Factor= Level.
4. The emission position marked as "-" means no suspected emission found with sufficient margin against limit line or noise floor only.